

Subhankar Satpathy

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SUMMARY

Full-stack developer (B.Tech '27) with production-grade project experience in real-time collaborative systems, LLM-powered tooling, and scalable backend architecture using TypeScript, React, Node.js, and PostgreSQL. Solved **600+ algorithmic problems** — LeetCode **1769 (Knight)**, Codeforces **Specialist (1459)**, CodeChef **3-star** (max 1669).

TECHNICAL SKILLS

Languages: C++, Java, TypeScript, JavaScript (ES6+), SQL

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3, SSR, WebRTC

Backend: Node.js, Express.js, REST APIs, WebSockets, JWT, OAuth, OpenAI API, Groq API, LLM Integration

Databases & DevOps: PostgreSQL, MongoDB, Redis, Prisma, pgvector, Docker, GitHub Actions, AWS (S3, EC2), Vercel, Jest, Turborepo, Linux

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, System Design

EDUCATION

Birla Institute of Technology, Mesra

B.Tech, Electronics and Communication Engineering

2023 – 2027

Current CGPA: 8.15 / 10

DAV Public School, Pokhariput

Senior Secondary (Class XII), CBSE

2022

Percentage: 95.4%

PROJECTS

Scriblio | *AI-Powered Collaborative Whiteboard* | [Live Demo](#) | [GitHub](#)

Sept 2025 – May 2026

Tech: TypeScript, React, Vite, Yjs (CRDT), y-webrtc, WebSockets, Node.js, Redis, OpenAI / Groq API, Docker, k6

- Built a real-time collaborative whiteboard with Yjs CRDTs, WebRTC P2P sync, and a custom WebSocket signaling server; load-tested the WebSocket signaling server with k6, where **25 concurrent clients** exchanged **~9,800 messages** over a **60s run** with **0 connection errors** and **107ms median round-trip latency** against the live cloud deployment.
- Designed a 4-service architecture with a React client, WebSocket signaling server, AI command service, and Redis presence layer, containerized with Docker Compose and using hybrid WebRTC + WebSocket transport for offline-tolerant P2P collaborative writes.
- Integrated a natural-language LLM command palette (Cmd+K) via OpenAI/Groq API to generate shapes, diagrams, and layouts from text prompts; in a **22-participant usability study**, median diagram-creation time dropped from **95s to 38s (60% reduction)**, with 82% of users reporting the AI flow as faster and a 4.3/5 “would use again” score.

Chess4Nerds | *Real-Time Multiplayer Chess Platform* | [Live Demo](#) | [GitHub](#)

Sept 2025 – May 2026

Tech: TypeScript, React.js, Node.js, Express.js, WebSockets, Redis, PostgreSQL, Prisma, Turborepo, GitHub Actions

- Built a real-time multiplayer chess platform sustaining **110+ concurrent matches** at **305 moves/sec** with **p95 112ms** move propagation under k6 load (220 VUs, 60s, single Node WS replica), with server-authoritative move validation via chess.js, async write-behind to Postgres, and DB-backed game rejoin on reconnect.
- Added an optional Redis pub/sub layer that relays game-room broadcasts across WebSocket replicas; shipped GitHub Actions CI running Prisma schema validation, WebSocket/backend builds, and the engine + rating test suites on every PR.
- Shipped Elo-band matchmaking (± 100 , widens with wait time) and a from-scratch **minimax + alpha-beta** AI opponent with MVV-LVA move ordering and piece-square-table evaluation, configurable at easy/medium/hard via search depth 2/3/4.

ReadHaven | *Social Book Discovery Platform* | [Live Demo](#) | [GitHub](#)

Jan 2026 – May 2026

Tech: Next.js 16, TypeScript, PostgreSQL (pgvector), Prisma, NextAuth, Tailwind CSS, Jest, GitHub Actions, Vercel

- Built a full-stack social reading platform (Next.js 16 App Router, SSR) with friends/activity feed, posts with likes and comments, shelves, reviews, yearly reading goals, and statistics dashboards over a **7K+ book catalog** seeded from Open Library with ISBN deduplication.
- Shipped **vector-similarity book recommendations** using pgvector cosine-distance with an ivfflat index over 384-dim sentence embeddings (all-MiniLM-L6-v2); exposed via a /discover semantic-search page and “More like this” on book detail pages, with composite (userId, createdAt) indexes on the reviews and posts tables for time-ordered per-user queries.
- Authored a Jest test suite with **91% line / 88% statement coverage** (132 tests across 19 suites on auth and API handlers) and GitHub Actions CI/CD running lint, typecheck, jest, and production build against an ephemeral pgvector/pgvector:pg16 service container; shipped full SEO with **Lighthouse 100 SEO / 90 Accessibility**.

ACHIEVEMENTS

- Solved **600+ algorithmic problems** across LeetCode, Codeforces, and CodeChef.
- **LeetCode** contest rating **1769 (Knight)**; **Codeforces Specialist** (max 1459); **CodeChef 3-star** (max 1669).
- **Rank 171** globally in CodeChef Starters 216 (Division 2).
- **Finalist, Smart India Hackathon (SIH) 2025** — national-level hackathon; shipped a full-stack product in a 36-hour sprint.
- **Winner, HackQuest 2025** — delivered a full-stack application end-to-end within a 24-hour sprint.

LEADERSHIP & EXTRACURRICULARS

- **Technical Lead, IEEE Student Chapter, BIT Mesra:** Led a 5-person dev team using Git-flow — reviewed **20+ PRs**, enforced code quality standards, and shipped **3 club projects** end-to-end; mentored juniors in web dev and competitive programming; organized hackathons and workshops for **80+ students**.
- **Honors:** NTSE Scholar and KVPY Fellow — national merit-based scholarships.
- Ranked **top 1%** globally on Chess.com (1750 Elo).